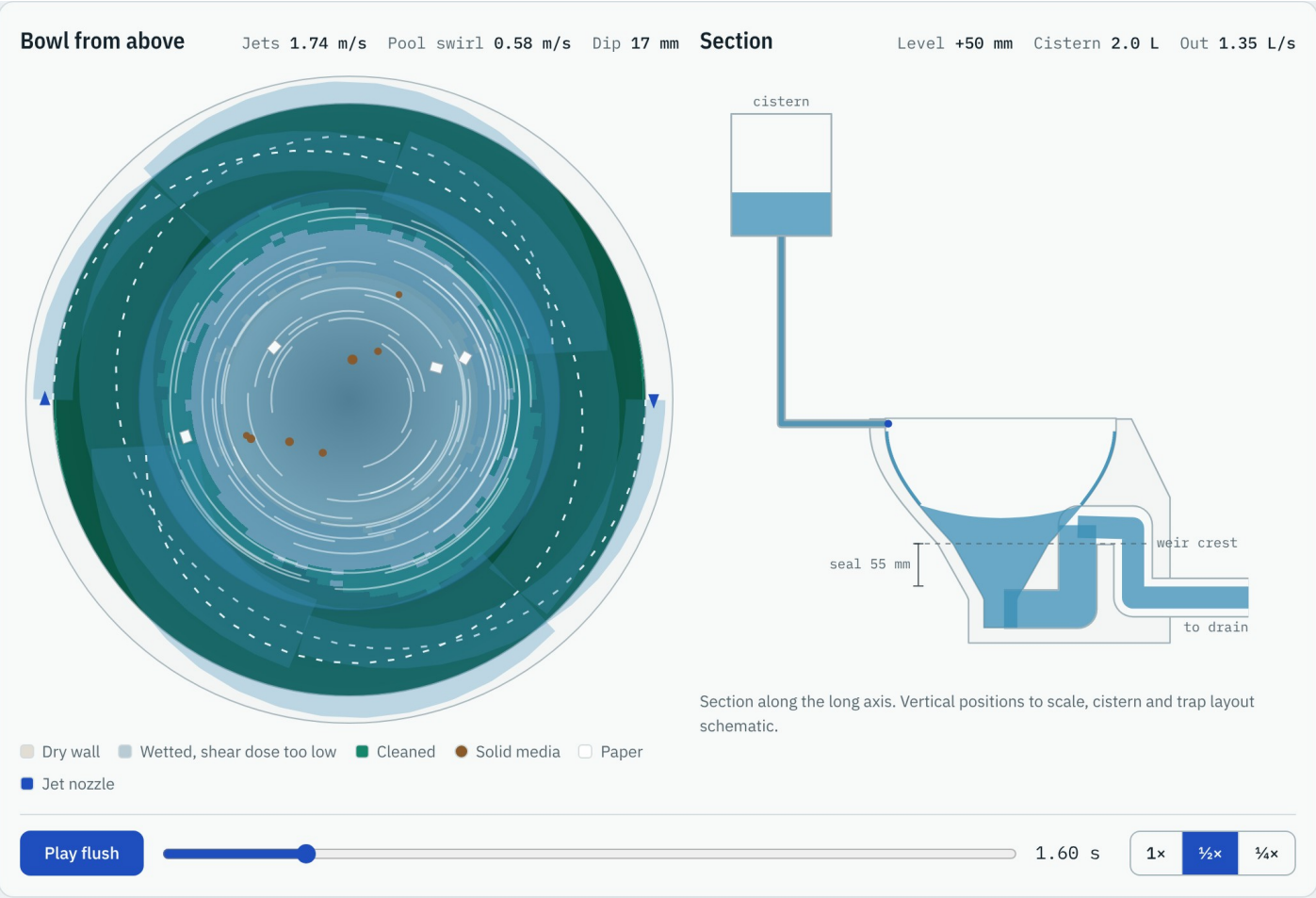
Tangential jet flush model

One flush in milliseconds: compare designs quickly, then check the best ones with CFD



ILLUSTRATIVE ESTIMATES, DEFAULT DESIGN

88 %

of the bowl wall above the waterline cleaned

99 %

of a 250 g solid test load removed

4.5 L

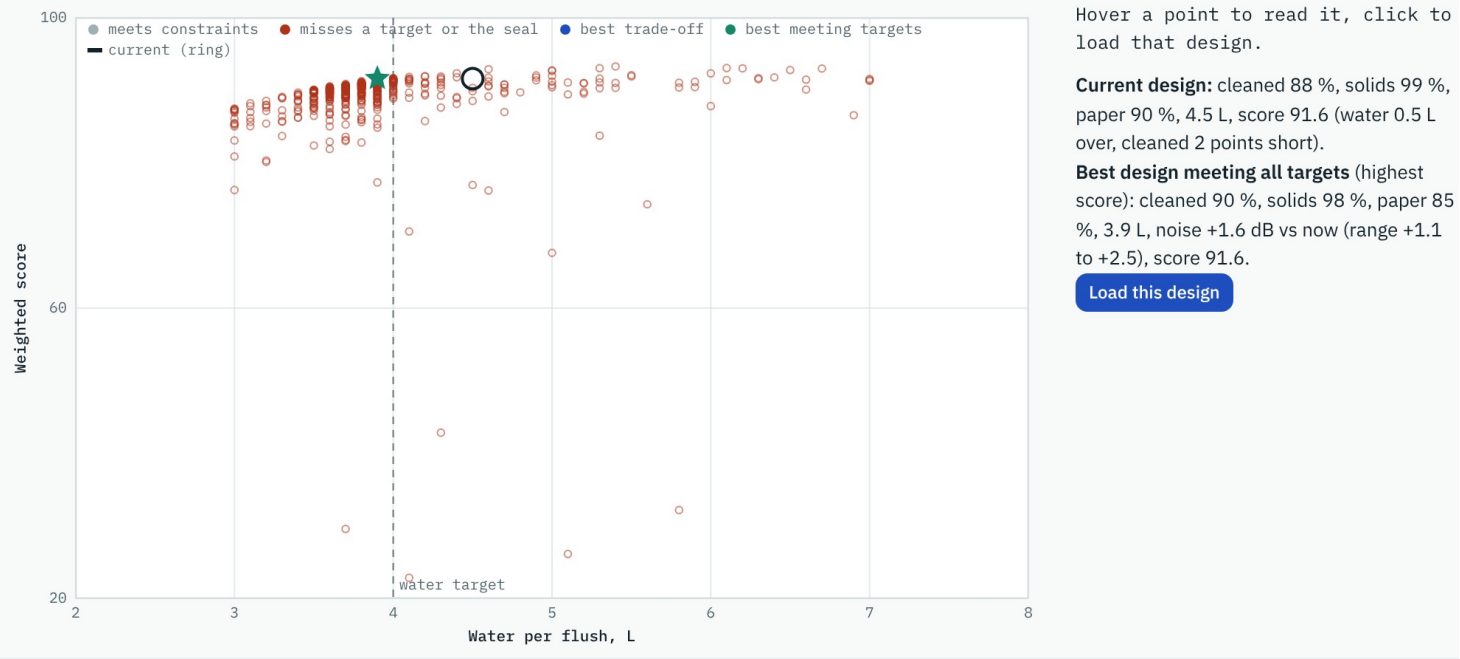
water per flush, EU washdown, full flush

Try it live
fintlp.github.io/jet-flush-model

Illustrative estimates from a reduced-order model. Coefficients are not yet calibrated against CFD or rig data, so results show trends between designs, not validated values.

AI lab: let the model search

Three ways AI works with the physics model, all running in the browser



Each dot is one simulated flush; red rings miss a target. The blue curve is the best trade-off between score and water, and the green star is the best design that meets all targets.

5–20 ms
physics model, per flush

≈ 5 µs
neural surrogate, per design

minutes to hours
PreonLab or CFD, rough estimate

1 Design optimiser

Finds the best settings you allow, or a design that meets targets such as at most 4.0 L, 90 % cleaned and no louder than now. About 540 flushes in seconds.

2 Neural surrogate

Learns the model from 300 flushes and answers roughly a thousand times faster. In practice it would be trained on PreonLab runs.

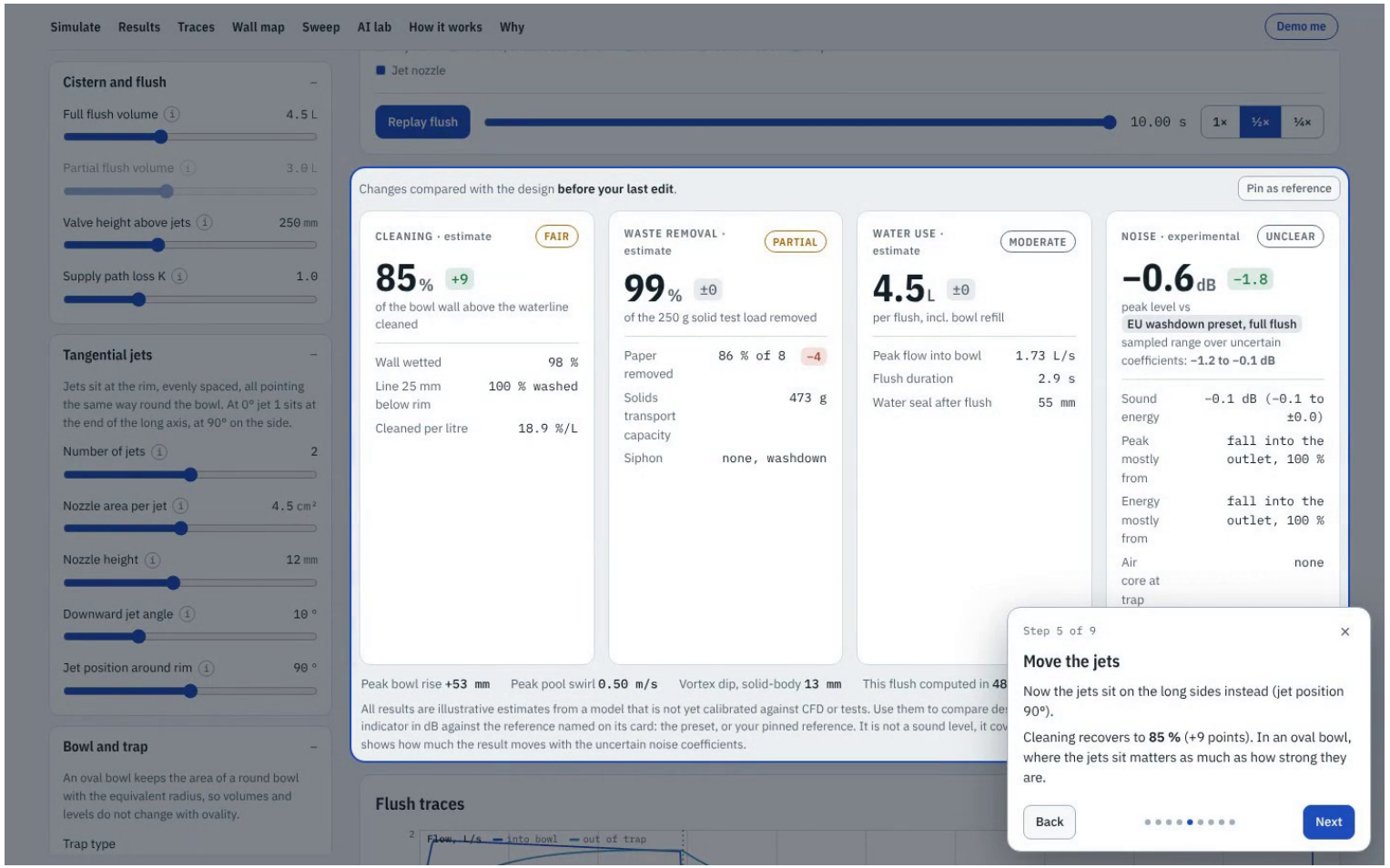
3 Hybrid calibration

Fits the model to reference runs and corrects the rest with data. On synthetic test data it cuts the error on unseen designs by about two to eight times.

Results come from the uncalibrated model: suggestions are candidates for a CFD or rig check, not final answers. The calibration demo uses synthetic reference data.

Guided demo in one minute

Play a flush, make the bowl oval, let the AI find a better design that is no louder, all live in the browser



88 % → 76 %

cleaning when the bowl turns oval, jets at the ends

76 % → 85 %

after moving the jets to the long sides

87 % at 4.0 L

AI lab design: more cleaning, 0.5 L less water, no louder

Start it yourself
fintlp.github.io/jet-flush-model/#demo

Recorded from the live page (v1.3.0) on 20 October 2026. Illustrative estimates from the uncalibrated model; noise is a relative, experimental indicator.